

Exercise and worked solution

Risk measures: VaR, ES & volatility

Four parts to master tail measures : compute VaR and ES in the usual cases, read a discrete case, show that VaR sometimes punishes diversification, and update the volatility that feeds them. Work through each part before reading the solution.

Part A · VaR and ES, normal case

PROBLEM

Losses follow $N(-20, 30)$ (in M). With $z_{99} = 2.326$ and $\phi(2.326) = 0.02665$, compute (a) the 99% VaR, (b) the 99% ES, and check the relation between them.

SOLUTION

- (a) $\text{VaR} = \mu + \sigma \cdot z = -20 + 30 \times 2.326 = \mathbf{49.79M}$.
- (b) $\text{ES} = \mu + \sigma \cdot \phi(z)/(1 - X) = -20 + 30 \times 0.02665/0.01 = -20 + 79.95 = \mathbf{59.96M}$. Indeed $\text{ES} \geq \text{VaR}$: ES averages losses all above the VaR. Trap: do not forget the mean term -20 , and divide $\phi(z)$ by $(1 - X) = 0.01$.

Part B · Discrete case

PROBLEM

A portfolio suffers losses of 2, 5 or 8M with probabilities 88%, 10% and 2%. Compute the 99% VaR, the 97% VaR and the 97% ES.

SOLUTION

- Cumulative probabilities: $[0-88] \rightarrow 2M$, $[88-98] \rightarrow 5M$, $[98-100] \rightarrow 8M$. 99% VaR: the 99 percentile falls in $[98; 100]$, so **8M**. 97% VaR: the 97 percentile falls in $[88; 98]$, so **5M**.
- 97% ES: the 3% tail holds 8M (2%) and 5M (1%), renormalised to $2/3$ and $1/3$. $\text{ES} = (2/3) \times 8 + (1/3) \times 5 = \mathbf{7M}$ > 97% VaR. At the exact 98% boundary, VaR is ambiguous and the convention takes the average (6.5M).

Part C · VaR punishes diversification

PROBLEM

Two independent insurance contracts: each suffers 20M with probability 0.8%, otherwise 1M. (a) 99% VaR of one contract? (b) 99% VaR of the portfolio of both? (c) Conclusion?

SOLUTION

- (a) The 20M loss (0.8%) is below the 1% threshold: it escapes the 99% VaR, which is only **1M**.
- (b) By independence: 2M (98.4064%), 21M (1.5872%), 40M (0.0064%). The probability of at least one big loss (1.5936%) exceeds 1%, so the 99 percentile falls on 21M: **99% VaR = 21M**.
- (c) $21 > 1 + 1$: the diversified portfolio's VaR exceeds the sum of the VaRs, though nothing changed economically. This is a subadditivity violation: VaR is not coherent. ES, by contrast, stays subadditive.

Part D · EWMA and GARCH volatility

PROBLEM

Yesterday's EWMA volatility is $\sigma_{n-1} = 1.5\%$ per day ($\lambda = 0.94$). The market moves by $r_{n-1} = 3\%$. (a) New EWMA volatility? (b) How does a GARCH(1,1) differ at long horizon?

SOLUTION

- (a) $\sigma_n^2 = 0.94 \times 0.015^2 + 0.06 \times 0.03^2 = 0.00021150 + 0.00005400 = 0.0002655$, so $\sigma_n = \mathbf{1.629\%}$. The shock lifted volatility from 1.5% to 1.63%.
- (b) EWMA lets this volatility persist with no anchor. GARCH(1,1) adds a weight on the long-run variance V_L ($\sigma_n^2 = \omega + \alpha r^2 + \beta \sigma_n^2$, $\omega = \gamma V_L$): its forecast returns gradually toward V_L (mean reversion), giving better long-horizon forecasts. If current volatility is above V_L , GARCH forecasts a decline.

Key takeaways: VaR = X percentile of losses (normal $\mu + \sigma z$, discrete by cumulated bands); ES = $E[L \mid L > \text{VaR}] \geq \text{VaR}$ (renormalised

tail average); VaR violates subadditivity while ES is coherent, hence FRTB $\rightarrow$ ES 97.5%; EWMA updates $\sigma^2 = \lambda\sigma^2 + (1-\lambda)r^2$ and GARCH adds a long-run anchor and mean reversion.